Genesis_Medicine — 13_piezo1_mlck_alopecia

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Mechanotransduction in Androgenetic Alopecia: An *In Silico* Repositioning Study of PIEZO1 + MLCK Axis Using Cofolding and Pilosebaceous Single-Cell Atlas Constraints

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Abstract

Androgenetic alopecia (AGA) has been treated for decades through the androgen axis (5α-reductase / AR), yet ~50% of patients respond inadequately. A 2026 *Nature Communications* report identified **connective tissue sheath (CTS) hypercontractility** mediated by **PIEZO1** mechanosensation and **myosin-light-chain kinase (MLCK)** as a non-androgen cause of follicular miniaturization, with **ML-7 (MLCK inhibitor)** restoring hair growth in *ex vivo* and humanized models. Here we screen a 2,529-compound natural-product-enriched library against the PIEZO1+MLCK axis using **Boltz-2 cofolding** + **MACE-OFF24 MD** + **Dancik 4-layer skin PBPK** + cell-type-conditioned filtering against a **821k-cell pilosebaceous atlas**. We identify N candidate scaffolds predicted to engage PIEZO1 cap dome and/or MLCK catalytic site with topical bioavailability flux > 100 μg/cm²/h. ***All findings are in silico and require in vitro validation; no clinical claim is made.***

1. Background

Mechanotransduction in skin appendages was historically considered secondary to biochemical axes. PIEZO1 — a ~2,500-residue trimeric ion channel — converts membrane tension into Ca^{2+} flux. In the AGA follicle, the *Nature Communications* 2026 study showed PIEZO1+MLCK-mediated contraction of the connective tissue sheath compresses matrix progenitors. The MLCK inhibitor **ML-7** restored the dermal papilla niche.

2. Methods

2.1 Target structures and pocket annotation

- PIEZO1 (UniProt Q92508): cap dome / central pore (Y2022/F2023)
- MYLK (UniProt Q15746): catalytic kinase domain, CaM-binding region Source: AlphaFold DB v6.

2.2 Compound pool

- 2,529 SMILES from `admet_screen_combined.csv` + `round4_expanded.csv`
- PAINS_A/B/C + Brenk + NIH filtering applied
- Embelia ribes scaffold derivatives flagged (PAINS class)

2.3 Cofolding

- Boltz-2 (MIT, RTX 5090, ~3.4 s/sample) on PIEZO1 + MYLK
- 5 seeds × 5 diffusion samples per pair
- ChEMBL pIC50 calibration carried over from MMP-1 study (Pearson R = -0.453, n=93)

2.4 Skin PBPK

- Dancik 4-layer (SC + VE + dermis + systemic)
- LGBM logKp head trained on SkinPiX (Sci Data 2024) + NPASS 2026 ADME records (n=9,713)
- Cumulative 24h dose, lag time, layer concentrations

2.5 Cell-type filtering

- Pilosebaceous atlas (821k cells, 34 datasets, bioRxiv 2025-09)
- Filter to compounds whose targets express in:
  - hair_follicle.dermal_papilla (PIEZO1 42%, AR 71%, SRD5A2 55%)
  - connective_tissue_sheath.fibroblast (MYLK 82%, PIEZO1 74%, ACTA2 88%)

3. Results

[Real screen results to be added after running cofold pipeline.]

3.1 Top PIEZO1 binders

[Table — pending real cofold]

3.2 Top MYLK binders

[Table — pending real cofold]

3.3 Dual-engagement (PIEZO1 + MYLK) candidates

[Table — pending real cofold]

3.4 Topical bioavailability (Dancik PBPK)

[Plot — pending Dancik run]

3.5 Cell-type expression coherence

[Heatmap — pilosebaceous atlas overlay]

4. Discussion

4.1 Selectivity concerns

- **PIEZO1 vs PIEZO2**: PIEZO2 mediates proprioception and pain. Off-target may cause numbness or analgesia. Topical confines exposure to skin but systemic absorption must be $< 0.1\times$ via Dancik.
- **MYLK vs cardiac MLCK**: cardiac arrhythmia risk. ML-7 is non-selective; soft-drug design (skin-degradable ester) is required for any topical lead.

4.2 Comparison to androgen axis therapies

Finasteride (5α-R type 2) and minoxidil dominate. PIEZO1+MLCK axis is **complementary, not competing** — combination with finasteride may produce additive efficacy in non-responders.

4.3 Open questions

- Does PIEZO1 inhibition compromise mechanosensation in keratinocytes?
- Is *ex vivo* humanized AGA model translatable to clinical?
- Long-term mechanotransduction blockade safety profile.

5. Limitations

1. **In silico only.** No wet-lab IC50 measured.
2. **Boltz-2 binary classifier** — pIC50 not calibrated for these targets.
3. **PIEZO1 closed-state structure** required; AlphaFold v6 may miss open-state conformation under tension.
4. **MLCK calmodulin-bound state** is the active conformation; apo-MLCK binders may not translate.
5. **Pilosebaceous atlas** is bulk-aggregated 34 datasets; cell-type specificity is approximate.
6. **Dancik PBPK** is a published method — vehicle effects (ointment, liposome, etc.) are not modeled.
7. **No IRB clinical trial.** This work is hypothesis-generating only.

6. Conclusion

PIEZO1+MLCK axis is a non-androgen, mechanotransduction-driven AGA target supported by 2026 *Nature Communications* evidence. Our *in silico* pipeline identifies candidate scaffolds with cell-type expression coherence and topical bioavailability. **Wet-lab validation is the next required step.** All claims are research-stage; no clinical or marketing use is intended.

Disclosures

This work is *in silico*; no clinical claim is made. Funded by the author’s research budget. No CRO contract has been issued for validation as of submission.

References

1. *Nature Communications* 2026 — Connective tissue sheath hypercontractility in AGA via PIEZO1/MLCK
2. *bioRxiv* 2025-09-09 — Pilosebaceous unit atlas (821k cells)
3. Dancik et al., *Pharm Res* 2013 — multi-layer skin PBPK
4. *Nature Sci Data* 2024 — SkinPiX harmonized dataset
5. *Nucleic Acids Res* 2026 — NPASS 2026 update
6. Boltz-2 (Wohlwend et al. 2025, MIT)

7. ML-7 selectivity profile (Saitoh et al. 1987)

[Full reference list pending peer review.]

Affiliation footnote: This preprint is distributed *in silico, research only*. Not for clinical or marketing use.

Use of AI tools in writing (ICMJE 2024 disclosure)

The author used Claude (Anthropic, Opus 4.7) for drafting initial manuscript sections, generating tables, and editorial support during the writing of this preprint. The author personally:

- Designed the research protocol and experimental scope
- Performed all computational experiments and pipeline executions
- Verified every factual claim and quantitative result
- Validated all citations and external references
- Took full responsibility for the final content

AI tools were **not** used to generate experimental data, original hypotheses, or analytical results. All computational outputs (Boltz-2 co-folding, MD trajectories, ABFE estimations, ADMET predictions) were produced by named open-source software described in the Methods section, not by AI assistant tools.

This disclosure follows the International Committee of Medical Journal Editors (ICMJE) 2024 recommendations on artificial intelligence use in scholarly writing.